Hybrid finite element and neural network methods for parametric PDEs

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MACS seminar
("Modélisation, Analyse et Calcul Scientifique")

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Context and motivation

Many-query contexts: real-time simulation, digital twins, optimization, uncertainty quantification

$\Rightarrow$ **Goal:** solve a parametric PDE accurately and quickly, for many parameters $\boldsymbol{\mu}$

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Our focus: the expressivity of the approximation spaces

existing enrichments are problem-dependent ⇒ two hybridizations: classical structure + neural expressivity

Two hybridizations

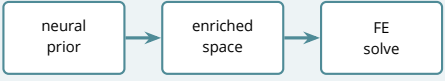
How does the neural network enter the approximation space V_h of a classical method?

Two hybridizations

How does the neural network enter the approximation space V_h of a classical method?

1st hybridization

Enriched FEM using PINNs



Network: pretrained, frozen

Guarantees: a priori error estimates

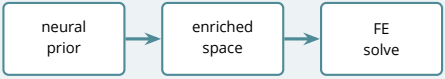
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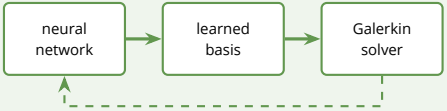
Network: pretrained, frozen

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Results: 1D, 2D, 3D, parametric

2nd hybridization

Learnable Galerkin methods



Network: learned through the solver

Guarantees: none (experimental)

Results: 1D, 3D, parametric

Parametric elliptic PDEs

Let $\Omega \subset \mathbb{R}^d$ be the spatial domain ($d = 1, 2, 3$) and $\mathcal{M} \subset \mathbb{R}^p$ the parameter space.
For one or several $\boldsymbol{\mu} \in \mathcal{M}$, find $u(\,\cdot\,;\boldsymbol{\mu}) : \Omega \to \mathbb{R}$ such that

$$\begin{cases} \mathcal{L}\big(u\,;\boldsymbol{x},\boldsymbol{\mu}\big) = f(\boldsymbol{x},\boldsymbol{\mu}), & (\boldsymbol{x},\boldsymbol{\mu}) \in \Omega \times \mathcal{M}, \\ \text{Dirichlet, Neumann or Robin BC}, & (\boldsymbol{x},\boldsymbol{\mu}) \in \Gamma \times \mathcal{M}, \end{cases} \tag{$\mathcal{P}$}$$

with $\Gamma := \partial\Omega$ the boundary of Ω, f the source term and $\mathcal{L}$ the linear differential operator defined by

$$\mathcal{L}(\,\cdot\,;\boldsymbol{x},\boldsymbol{\mu}) : u \mapsto R(\boldsymbol{x},\boldsymbol{\mu})\,u + \boldsymbol{C}(\boldsymbol{x},\boldsymbol{\mu}) \cdot \nabla u - \frac{1}{\mathrm{Pe}} \nabla \cdot \big(D(\boldsymbol{x},\boldsymbol{\mu}) \nabla u\big).$$

where R is the reaction, $\boldsymbol{C}$ the convection, D the (symmetric positive definite) diffusion matrix and Pe the Péclet number.

Numerical methods

Numerical methods for parametric PDEs*

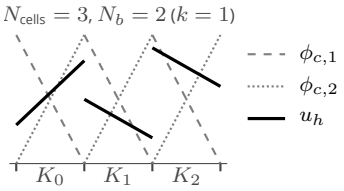
* presented with homogeneous Dirichlet boundary conditions

Discontinuous Galerkin (DG) method[1]

For a fixed $\boldsymbol{\mu} \in \mathcal{M}$, let $\mathcal{T}_h$ be a mesh of Ω of cells $(K_c)_{0 \leq c < N_{\text{cells}}}$.

Let V_h^{disc} be a discontinuous space of piecewise polynomials of degree k on the mesh $\mathcal{T}_h$. Its basis functions $(\phi_{c,i})_{c,i}$, with $\phi_{c,i}$ supported on a single cell K_c, decompose the discrete solution:

$$u_h(\boldsymbol{x}, \boldsymbol{\mu}) = \sum_{c,i} \beta_{c,i}(\boldsymbol{\mu})\, \phi_{c,i}(\boldsymbol{x}).$$



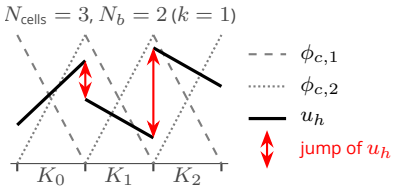
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Coupling the cells: ambiguous traces on the interfaces ⇒ numerical flux (it defines the method)

Diffusion-dominated regimes: flux for the diffusion term only, here SIPG (Symmetric Interior Penalty Galerkin)

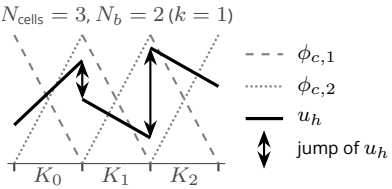
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Variational problem (SIPG):

Find $u_h \in V_h^{\text{disc}}$ such that, $\forall v_h \in V_h^{\text{disc}}$, $a_h^{\text{disc}}(u_h, v_h; \boldsymbol{\mu}) = l(v_h; \boldsymbol{\mu}) = \int_\Omega f\, v_h$, with

$$a_h^{\text{disc}}(u_h, v_h; \boldsymbol{\mu}) = \underbrace{\int_\Omega \left(R\, u_h v_h + \boldsymbol{C} \cdot \nabla_h u_h\, v_h + \frac{1}{\text{Pe}} D \nabla_h u_h \cdot \nabla_h v_h \right)}_{\text{cell terms}} \underbrace{- \frac{1}{\text{Pe}} \int_{\mathcal{F}_h} \left([\![u_h]\!] \cdot \{ D \nabla_h v_h \} + \{ D \nabla_h u_h \} \cdot [\![v_h]\!] \right)}_{\text{consistency}} \underbrace{+ \frac{1}{\text{Pe}} \int_{\mathcal{F}_h} \kappa\, [\![u_h]\!] \cdot [\![v_h]\!]}_{\text{penalty}}$$

where $\mathcal{F}_h$ is the set of faces, ∇_h the broken gradient, $[\![\cdot]\!]$ the jump, $\{\cdot\}$ the average, κ the penalty parameter.

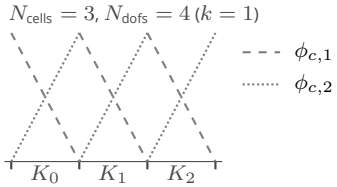
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Continuous Galerkin (CG) method[1]

Same mesh $\mathcal{T}_h$ and fixed $\boldsymbol{\mu}$ as DG.

Let $V_h := V_h^{\text{disc}} \cap C^0(\Omega)$ be a continuous space of piecewise polynomials of degree k on the mesh $\mathcal{T}_h$. Its basis functions $(\phi_I)_I$, with ϕ_I supported on the cells sharing the dof I, decompose the discrete solution:

$$u_h(\boldsymbol{x}, \boldsymbol{\mu}) = \sum_{I=1}^{N_{\text{dofs}}} \beta_I(\boldsymbol{\mu})\, \phi_I(\boldsymbol{x}).$$



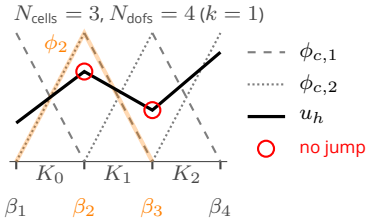
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Coupling the cells: shared dofs via the connectivity table ⇒ no jumps, no numerical flux

$$\text{conn} : (c, i) \mapsto I, \quad \beta_I(\boldsymbol{\mu}) = \beta_{c,i}(\boldsymbol{\mu}), \quad \phi_I = \sum_{\text{conn}(c,i)=I} \phi_{c,i}$$

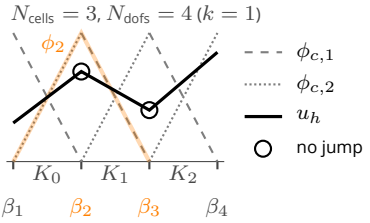
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with $V_h^0 \subset V_h$ vanishing on Γ and no interface terms, no penalty.

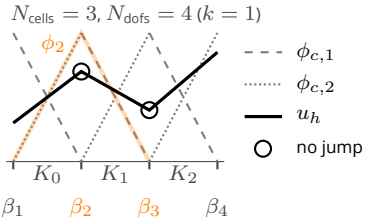
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Linear system (in both Galerkin methods): $A_h(\boldsymbol{\mu})\, \boldsymbol{\beta}_h(\boldsymbol{\mu}) = \boldsymbol{b}_h(\boldsymbol{\mu})$

[1] [Ern and Guermond, 2004]

Physics-informed neural networks (PINNs)[1]

Let $u_\theta(\boldsymbol{x}, \boldsymbol{\mu})$ be a neural network with weights θ and inputs $(\boldsymbol{x}, \boldsymbol{\mu}) \in \Omega \times \mathcal{M}$, approximating the solution u.

Training: Find the weights $\theta^\star$ such that

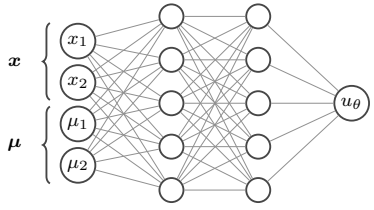
$$\theta^\star = \arg\min_\theta \big(\omega_r \, \mathcal{J}_r(\theta) + \omega_b \, \mathcal{J}_b(\theta)\big), \qquad (\mathcal{P}_\theta)$$

with ω_r and ω_b the weights of the two loss terms, and

residual loss
$$\mathcal{J}_r(\theta) = \int_{\mathcal{M}} \int_{\Omega} \big|\mathcal{L}\big(u_\theta(\boldsymbol{x}, \boldsymbol{\mu}); \boldsymbol{x}, \boldsymbol{\mu}\big) - f(\boldsymbol{x}, \boldsymbol{\mu})\big|^2 \, \mathrm{d}\boldsymbol{x} \, \mathrm{d}\boldsymbol{\mu},$$

boundary loss
$$\mathcal{J}_b(\theta) = \int_{\mathcal{M}} \int_{\Gamma} \big|u_\theta(\boldsymbol{x}, \boldsymbol{\mu})\big|^2 \, \mathrm{d}\boldsymbol{x} \, \mathrm{d}\boldsymbol{\mu}.$$

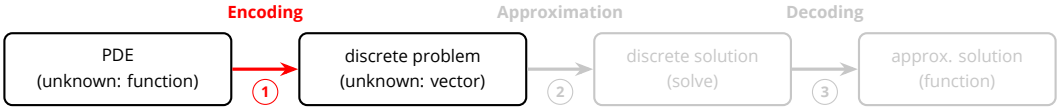
Monte-Carlo method:
Discretize the cost functions by random process.



[1] [Raissi, Perdikaris, and Karniadakis, 2019]

Link between Galerkin and PINNs

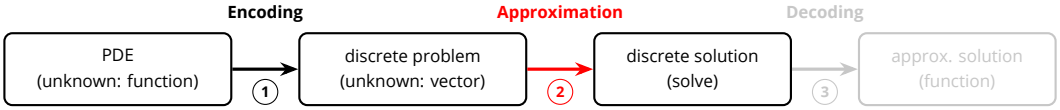
Numerical method (three steps):



		Galerkin (DG/CG)	PINN
	Space	vector space (V_h^{disc}/V_h)	manifold of the u_θ
1	Unknowns	dofs $\boldsymbol{\beta}_h(\boldsymbol{\mu})$	weights θ
	Integrals	quadrature on cells	Monte Carlo on collocation points

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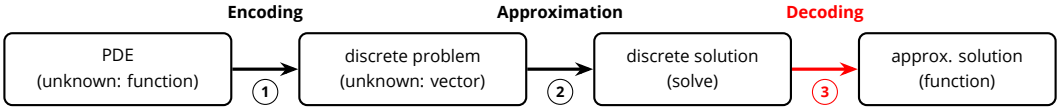
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②	Parameters	fixed $\boldsymbol{\mu}$, one solve per $\boldsymbol{\mu}$	$\boldsymbol{\mu} \in \mathcal{M}$ as input, one training
	Resolution	linear system $A_h(\boldsymbol{\mu})\boldsymbol{\beta}_h(\boldsymbol{\mu}) = \boldsymbol{b}_h(\boldsymbol{\mu})$	optimization on θ

Link between Galerkin and PINNs

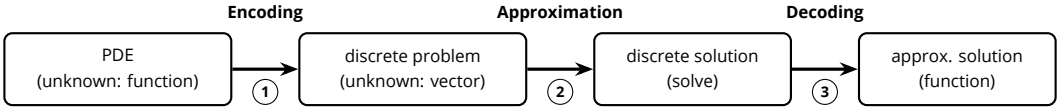
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③	Solution	$u_h(\boldsymbol{x}, \boldsymbol{\mu})$, linear in $\boldsymbol{\beta}_h$	$u_\theta(\boldsymbol{x}, \boldsymbol{\mu})$, non-linear in θ

Link between Galerkin and PINNs

Numerical method (three steps): **Same philosophy, different approximation space**



		Galerkin (DG/CG)	PINN
1	Space	vector space (V_h^{disc}/V_h)	manifold of the u_θ
	Unknowns	dofs $\boldsymbol{\beta}_h(\boldsymbol{\mu})$	weights θ
	Integrals	quadrature on cells	Monte Carlo on collocation points
2	Parameters	fixed $\boldsymbol{\mu}$, one solve per $\boldsymbol{\mu}$	$\boldsymbol{\mu} \in \mathcal{M}$ as input, one training
	Resolution	linear system $A_h(\boldsymbol{\mu})\boldsymbol{\beta}_h(\boldsymbol{\mu}) = \boldsymbol{b}_h(\boldsymbol{\mu})$	optimization on θ
3	Solution	$u_h(\boldsymbol{x}, \boldsymbol{\mu})$, linear in $\boldsymbol{\beta}_h$	$u_\theta(\boldsymbol{x}, \boldsymbol{\mu})$, non-linear in θ

1st hybridization

Enriched finite element method using PINNs

Presentation

Pipeline of the hybrid method

TODO

Additive approach

TODO

Convergence analysis

TODO

Numerical results

Test case 1
Test case 2

Numerical results

Test case 1

Test case 2

Test case 1

TODO

Numerical results

Test case 1

Test case 2

Test case 2

TODO

Conclusion, limitations and perspectives

Conclusion, limitations and perspectives

TODO

2nd hybridization

Learnable Galerkin methods

Introduction

Autodiff

TODO

Formalization

Optimization problem

TODO

Non-parametric enrichment (PINN loss)

Algorithm

TODO

Parametric enrichment (neural operator)

Algorithm

TODO

Conclusion

Conclusion

Contribution 1 : TODO

TODO

Contribution 2 : TODO

TODO

Perspectives :

- TODO

Appendix – 1st hybridization

Enriched finite element method using PINNs

Additional details

Additional details

TODO

Détails de la méthode (hors présentation principale)

Additional numerical results

Additional numerical results

TODO

Résultats numériques supplémentaires

Appendix – 2nd hybridization

Learnable Galerkin methods

Additional details

Additional details

TODO

Détails de la méthode (hors présentation principale)

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Résultats numériques supplémentaires